

Deep Learning Hint Agent: A Next-Action Decision Framework for Progressive Competitive Programming Hints

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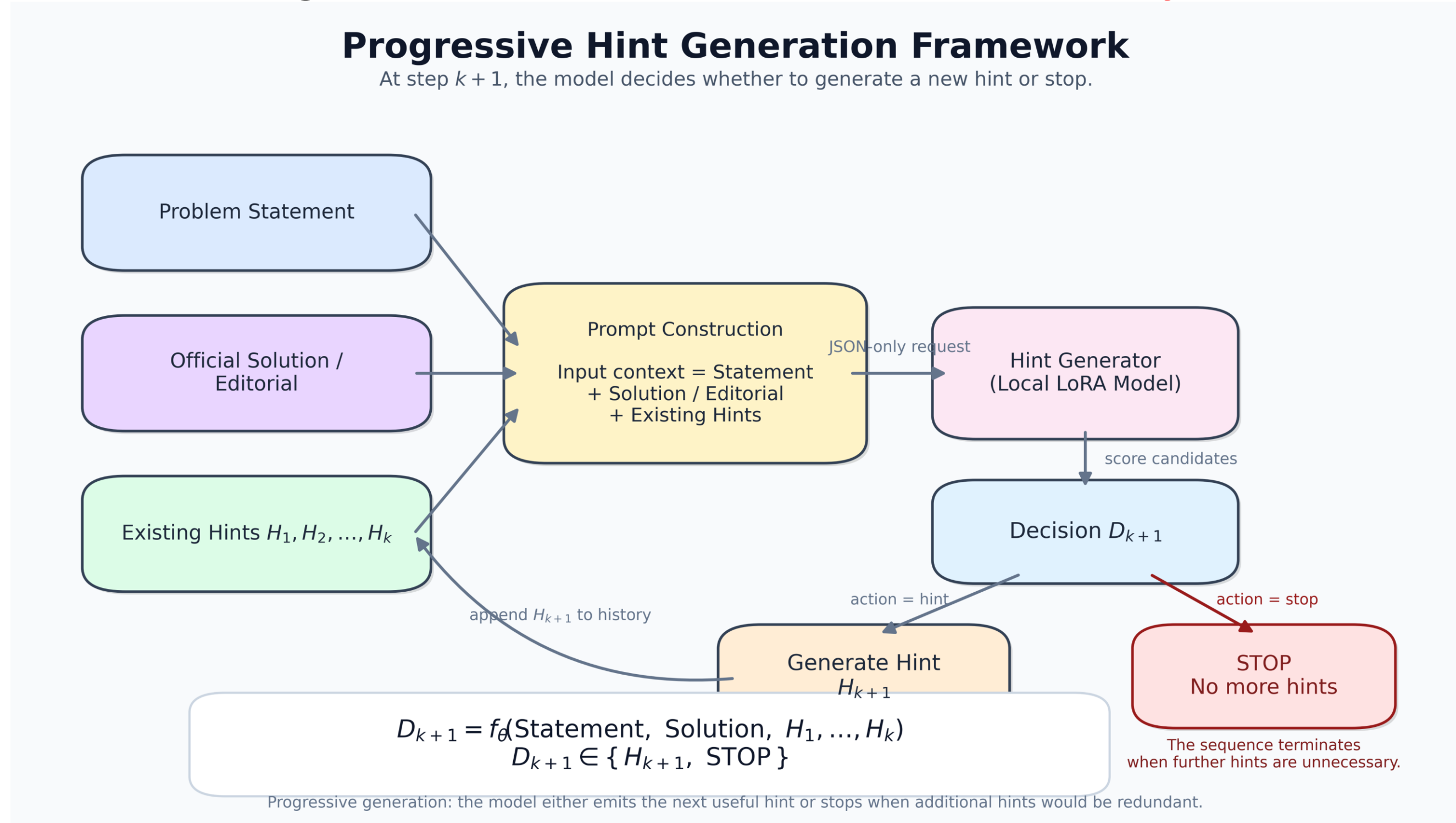
1. Abstract & Introduction

Problem:

In competitive programming (e.g., Codeforces), general LLMs in a zero-shot setting fail to provide adaptive hints. They either leak full solutions prematurely or provide descriptive, static text that undermines the pedagogical goal of provoking human critical thinking.

Our Core Idea:

Instead of modeling the multi-turn hint dialogue as a single continuous sequence (which forces the model to learn a statistical average length), we reformulate hint generation as a **Next-Hint Decision Policy**.



2. Data Collection & Processing

Multi-Source Pipeline: Overcame Codeforces Cloudflare protection by scraping contest metadata, blog editorials, and Luogu mirror statements, using automated retry loops to eliminate dynamic rendering blocks.

Dataset & Alignment: Structured data into unified statement, hints, and solutions payloads, and engineered custom data typologies

(**raw_hint_decision**, **synthetic_early_stop_rejection**, **synthetic_repeat_rejection**) for **robust SFT alignment**.

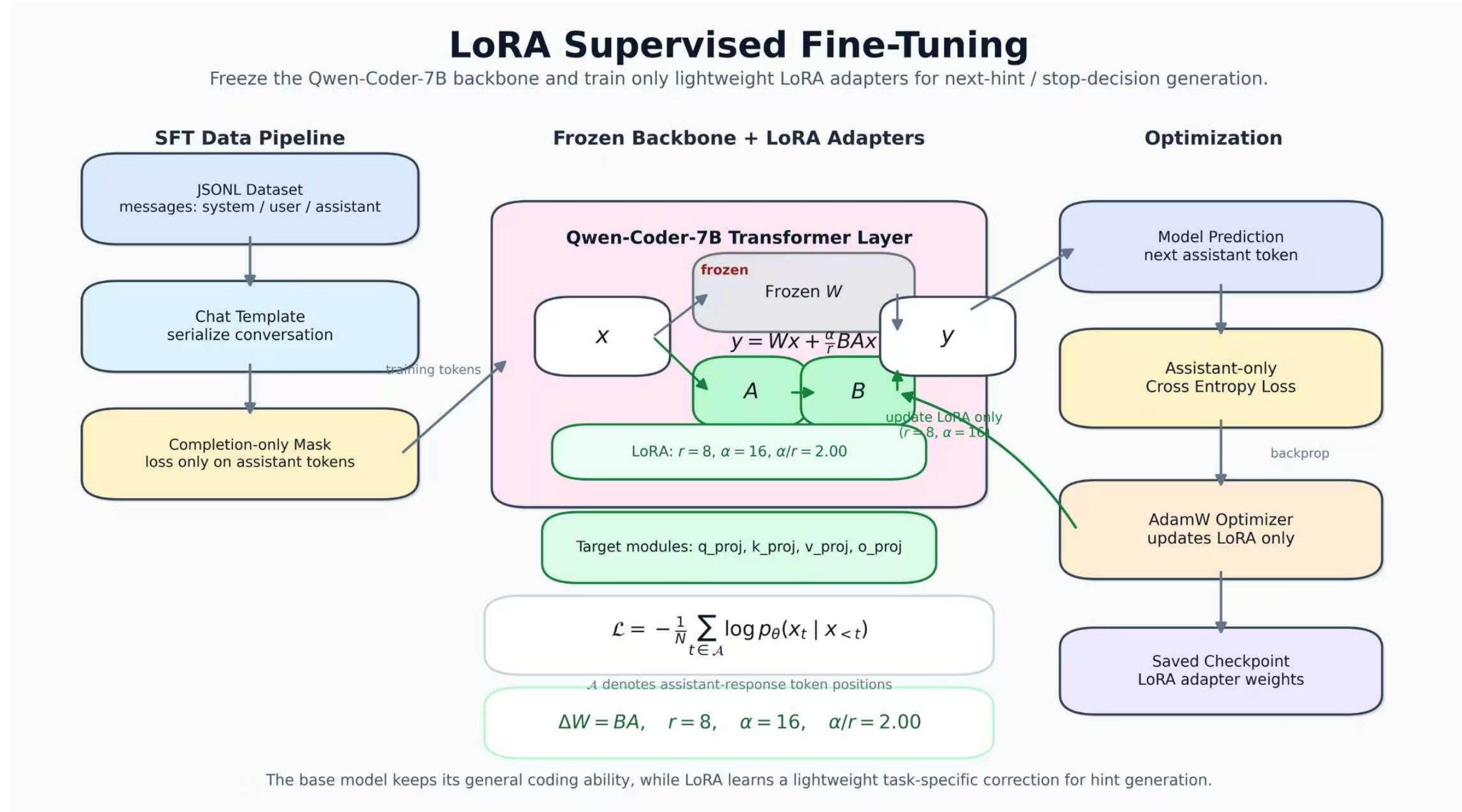
3. Methodology & System Improvements

Instruction Alignment: Implemented a short-epoch SFT strategy (150~250 epochs) as an implicit regularizer to preserve linguistic variation and prevent over-fitting.

Negative Rejection Data:

Expanded the training distribution with synthetic negative samples (**early_stop_rejection**, **repeat_rejection**) to penalize premature cutoffs and circular reasoning.

Inference Control: Replaced greedy decoding with a multi-candidate pipeline that spawns 10 parallel paths, leveraging a heuristic evaluation layer with Jaccard similarity scoring to filter and penalize semantic near-duplicates.



4. Empirical Results & Analysis

4.1. Zero-Shot Multi-Model Benchmarks

The Static Fallacy: Commercial models (Gemini-3.1, DeepSeek V3, Qwen3.6, Kimi K2.5 Pro) perform poorly on problems across Ratings 800–3300.

Model	Hint Count	Hint Length	Direct Reveal (Formulas)	Granularity & Detail	Scalability to Difficulty	Socratic Tone	Progressive Logic
Gemini 3.1 Flash Lite	4 (Low)	Excessively Long	Moderate	Overly Descriptive	Static	Low	Moderate
Qwen 3.6-35B-A3B	3 (Very Low)	Moderate	Low	Too Abstract/General	Static	Moderate	Moderate
DeepSeek V3	6 (Moderate)	Long	High (Frequent)	Instructional	Static	Low	High
Kimi K2.5 Pro	12 (Excessive)	Short	Moderate	Overly Granular	Static	Low	High

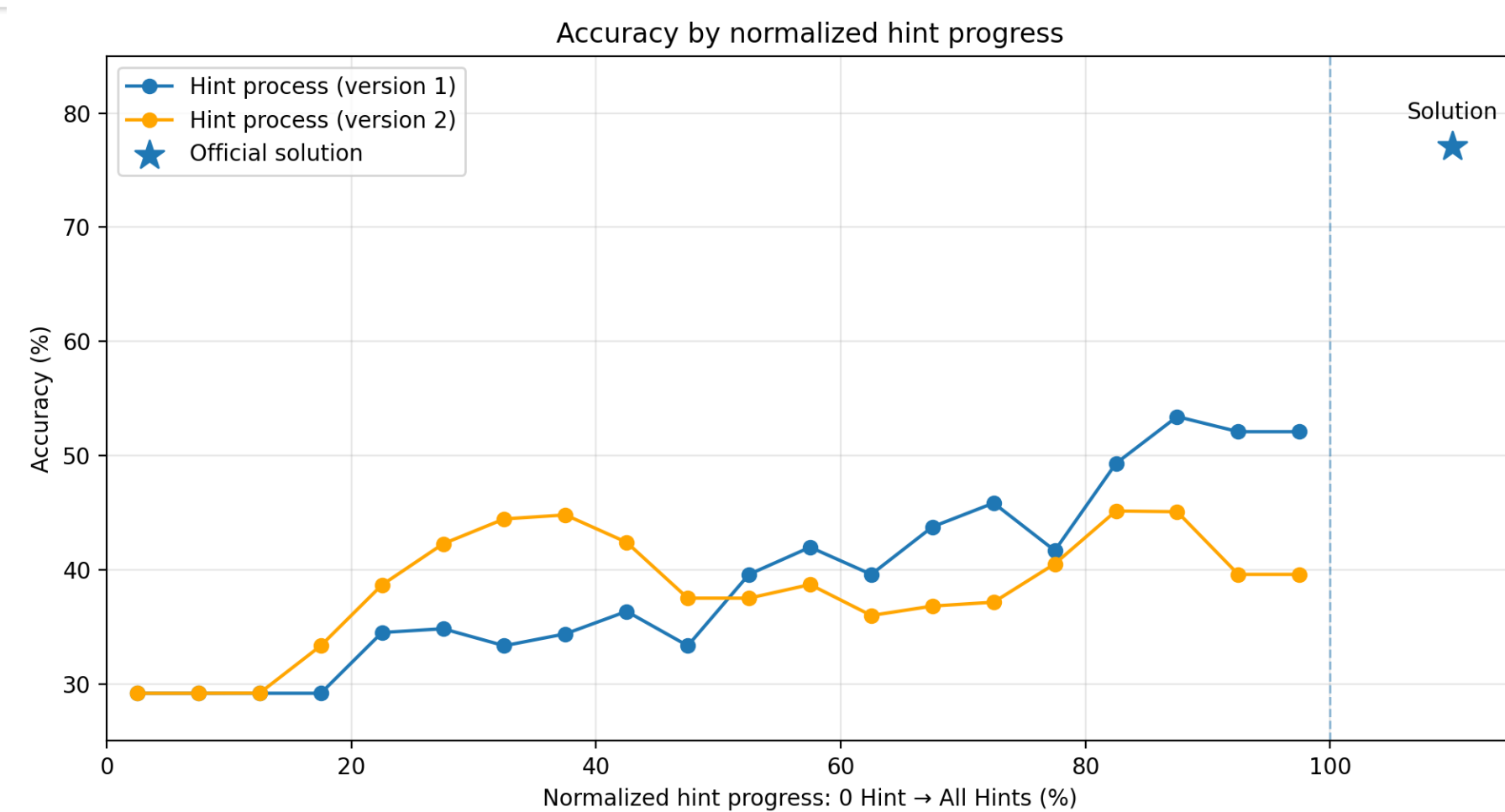
4.2. External Solver

Evaluation Loop

Automated Sandbox:

Deployed **DeepSeek-V4-Pro**

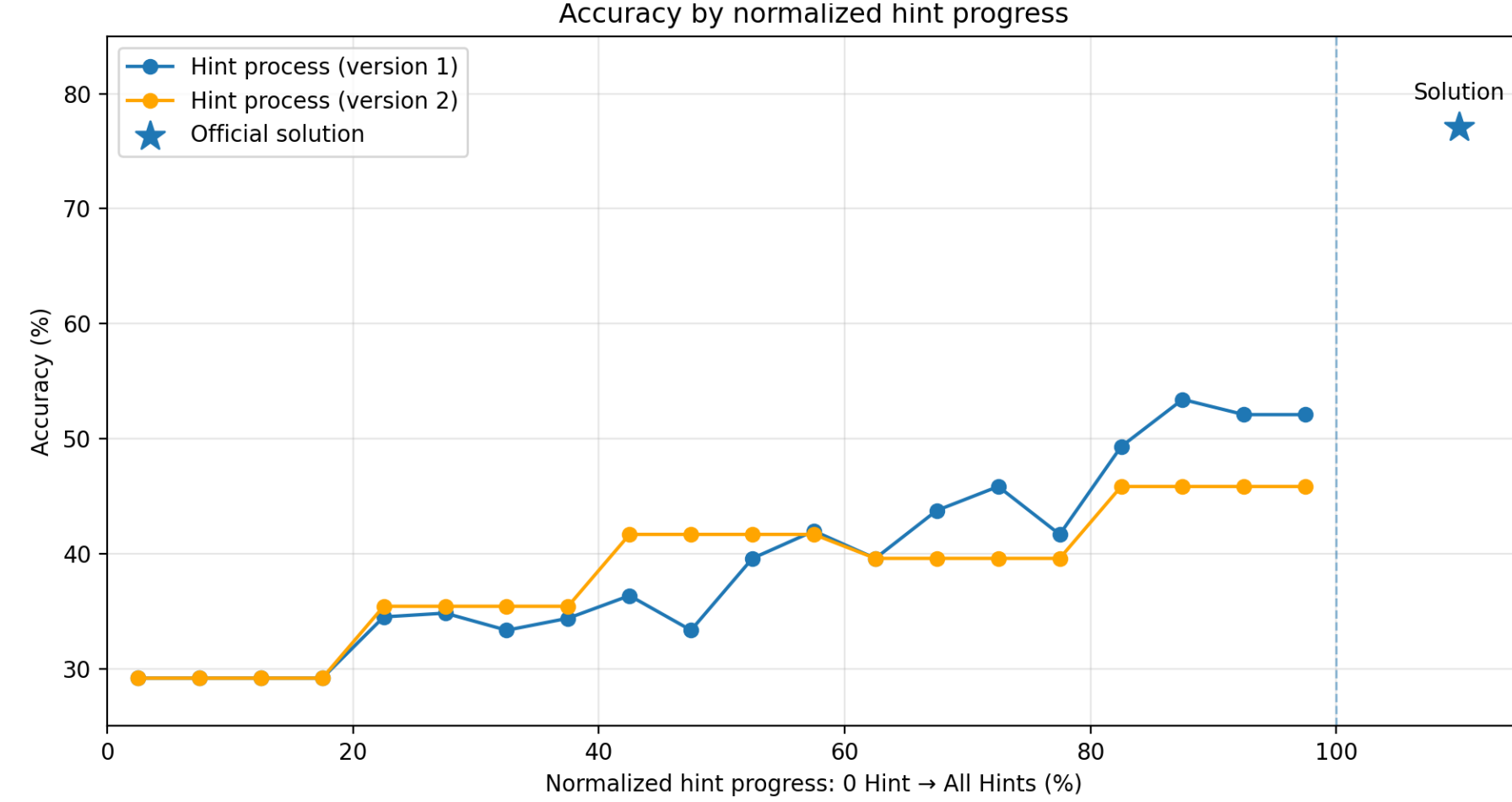
as a proxy user to write C++17 codes under sequential hint trajectories.



Trajectory Tracking:

Progress is tracked via multi-bit tracking strings from 0 Hint to the solution upper-bound.

(e.g., 000111 means starting from the 3rd hint, **Deepseek** can write **AC** code.).



4.3. Fine-Tuned Agent Empirical Curves

Output Distribution (Fig 1): Version 2 (**No solution hint generation**) faces context limits in deep layers. Implementing a Hard Truncation (Generally, no effective hints can be given after step 4) creates a strange statistical curve.

Solver Accuracy (Fig 2):Version 1 (**With solution hint generation**):

Accuracy climbs approximately monotonically from **29.2%** to **53.3%** (Solution test: **77.1%**), validating coherent pedagogical scaffolding.Version 2 (Capped at 4): Establishes a stable performance plateau at **45.8%**, preserving early-stage precision.

5. Limitations & Future Directions

Evaluation Alignment: A mismatch exists between the high-level conceptual hints generated by our agent and the engineering execution bottleneck (syntax/compilation) of the proxy solver.

Generalization Paradigm: Demonstrates a resource-efficient framework where specialized small language models act as "**cognitive scaffolding**" to unlock the latent reasoning of larger, un-tuned general foundation models **without prohibitive full-scale training costs**.

6. Demo

We have developed an interactive UI demo that allows users to select from three distinct profiles of hint models.

The interface supports progressive, step-by-step hint generation on demand.

Furthermore, users can leverage an integrated DeepSeek solver to synthesize execution-ready code directly from the accumulated hint sequence.

3. Generate Code

DeepSeek from current hints

DeepSeek from solution

2. Choose Hint Model

Mode

✓ Balanced: checkpoint-175

Enthusiastic: checkpoint-125

Silent: checkpoint-225

Generated Hints

****Hint 1****
If \$\$\$x \ge 2\$\$\$, find minimum \$\$\$y\$\$\$ such that \$\$\$y \ge \text{ceil}(\frac{x}{2})\$\$\$.

****Hint 2****
Divide all elements in range \$\$\$[y+1, x]\$\$\$ on \$\$\$x\$\$\$ and then divide \$\$\$x\$\$\$ on \$\$\$y\$\$\$.

****Hint 3****
Recursively solve for \$\$\$[1, y]\$\$\$.

****Hint 4****
What is the complexity?